

Create detached PGP signatures for all files in a directory

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Iterates over every file in a given directory and creates a detached PGP signature (.sig) for each one using GnuPG.

Language

bash

Category

cryptography

Command

```
bash -c 'dir=$1; shift; for f in "$dir"/*; do [ -f "$f" ] &&
    gpg --detach-sign "$f"; done' _
```

Explanation

This oneliner uses a `bash -c` subshell to accept the target directory as a trailing argument. The variable `dir` receives `$1` (the directory path), then `shift` removes it so remaining arguments are available via `$@`. A `for` loop iterates over all entries matching `"$dir"/*`, and `[ -f "$f" ]` ensures only regular files are processed (skipping subdirectories, symlinks, etc.). For each file, `gpg --detach-sign` creates a separate `.sig` file containing the PGP signature. The `_` placeholder serves as `$0` inside the subshell, ensuring `$1` maps to the first real argument.

Tags

gpg, pgp, cryptography, signature, detach-sign, bulk, verification

Dependencies

gnupg

Arguments

1. **DIR** (Optional): Path to the directory whose files will be signed.
Default: .

Examples

1. `bash -c 'dir=$1; shift; for f in "$dir"/*; do [ -f "$f" ] && gpg --detach-sign "$f"; done' _ /path/to/releases` - Sign all files in /path/to/releases
2. `bash -c 'dir=$1; shift; for f in "$dir"/*; do [ -f "$f" ] && gpg --detach-sign "$f"; done' _ .` - Sign all files in the current directory
3. `alias bsign="bash -c 'dir=\{$1; shift; for f in \{"\{$dir\}"/*; do [ -f \{"\{$f\} " ] && gpg --detach-sign \{"\{$f\}"; done' _"` - Create an alias named 'bsign' for easy bulk signing
4. `bsign /path/to/releases` - Use the alias to sign all files in a directory

Output

```
file1.tar.gz.sig  
file2.iso.sig  
file3.deb.sig
```

Notes

- Each original file gets a corresponding .sig file in the same directory
- GPG will prompt for passphrase if the private key is passphrase-protected
- Uses the default signing key; set GPGKEY env var or use `--default-key` to specify another
- The `[ -f "$f" ]` check skips subdirectories and non-regular files
- Hidden files (dotfiles) are not matched by the glob `*`; use `.*` as well if needed
- For large numbers of files, consider using `gpg --batch --yes` to skip confirmation prompts

Warnings

- Existing .sig files will be silently overwritten without confirmation
- Ensure you have sufficient disk space for the signature files
- If GPG cannot access your private key, the loop will fail for every file
- Signatures are created in the same directory as the source files

See Also

- find-large-files-recursive

Status

reviewed

Safety

caution

Shell

bash

Platforms

linux, gnu-linux, freebsd, openbsd, netbsd

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